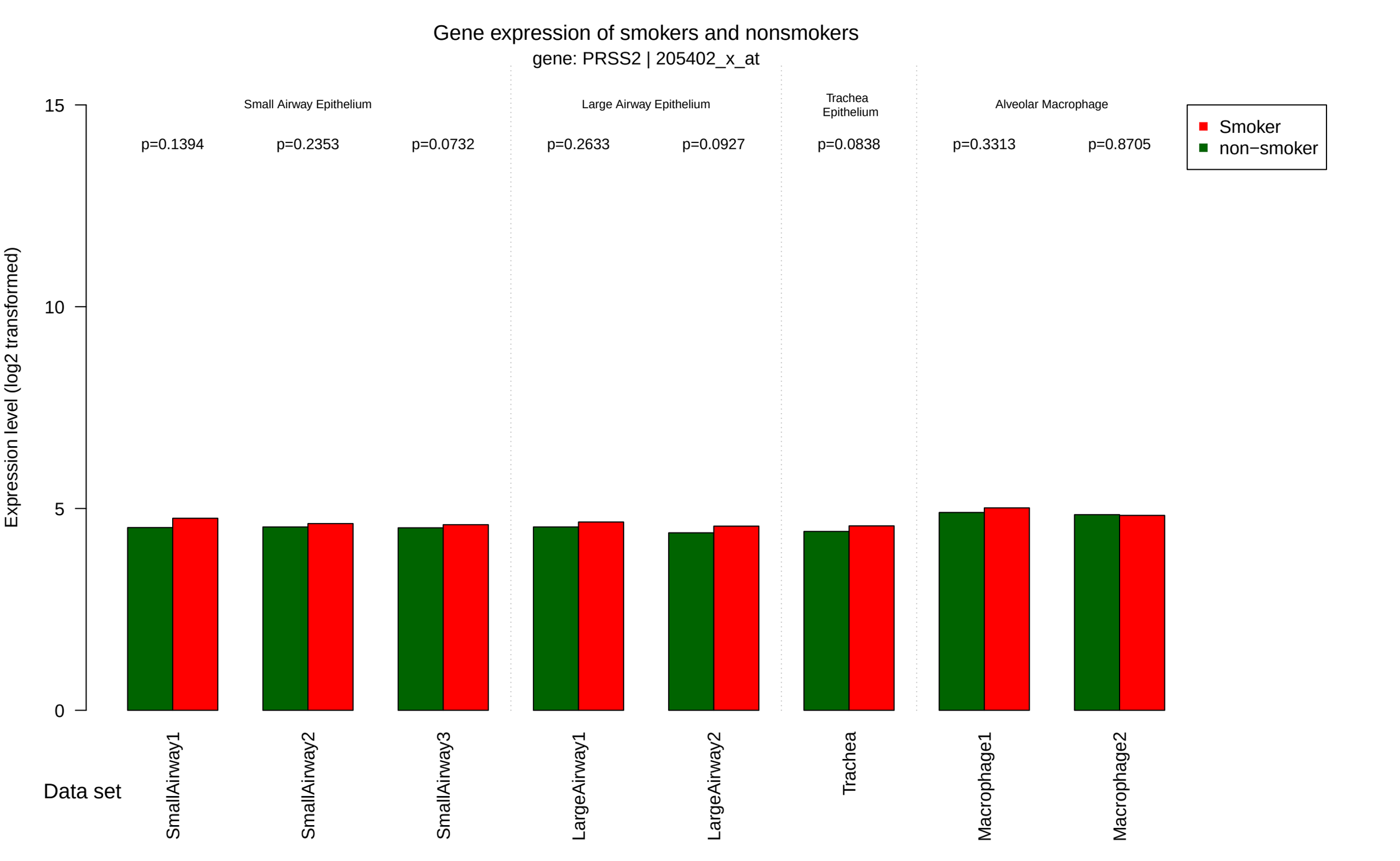
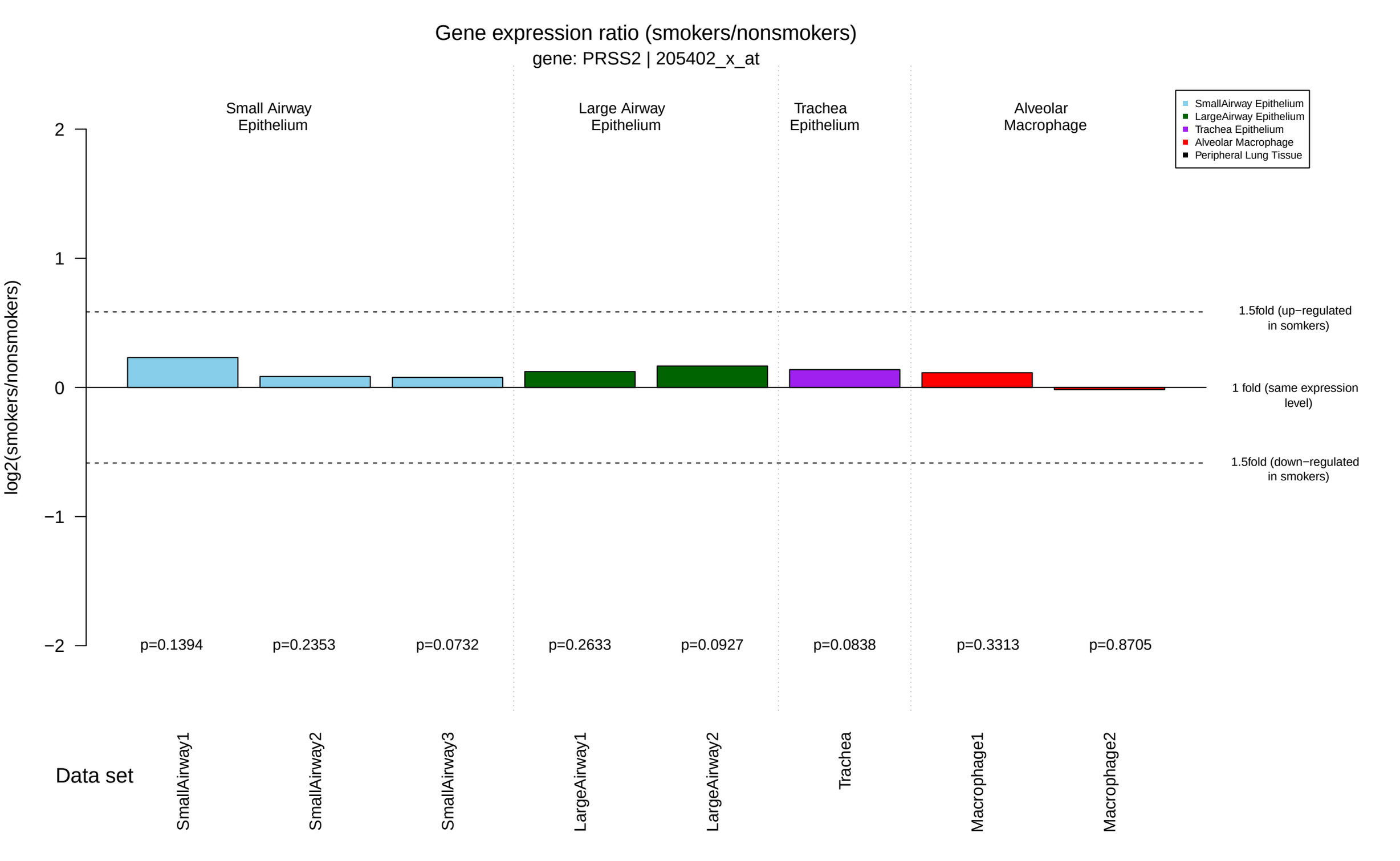
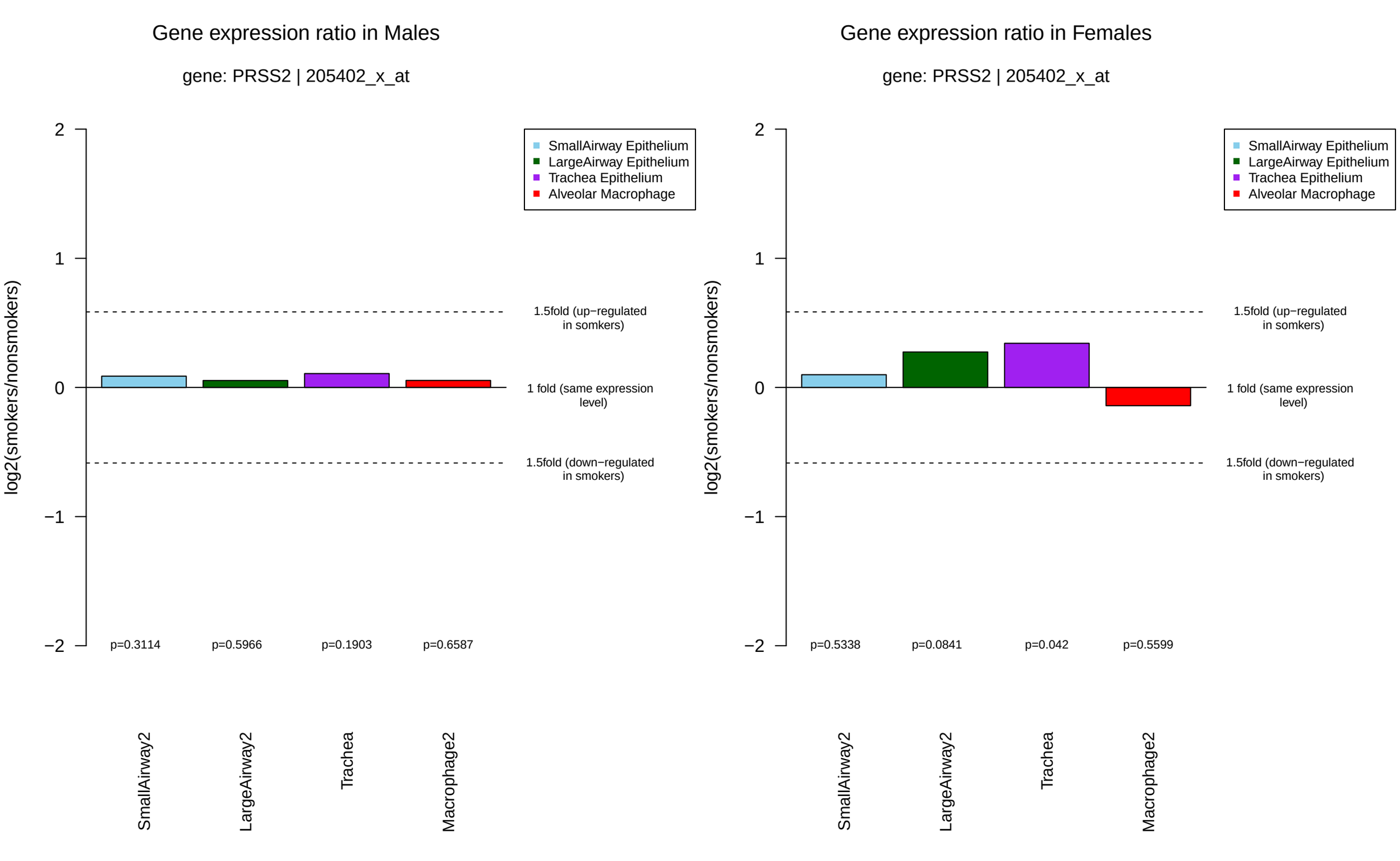
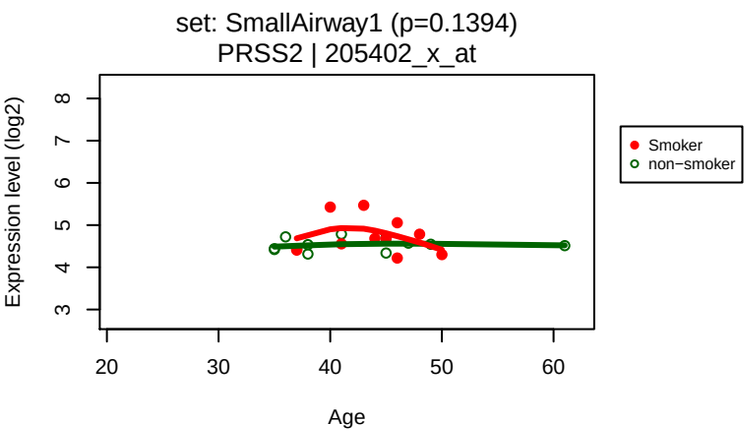
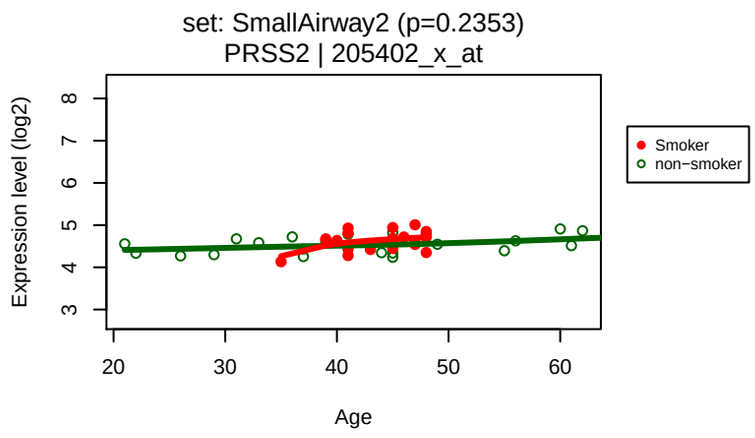
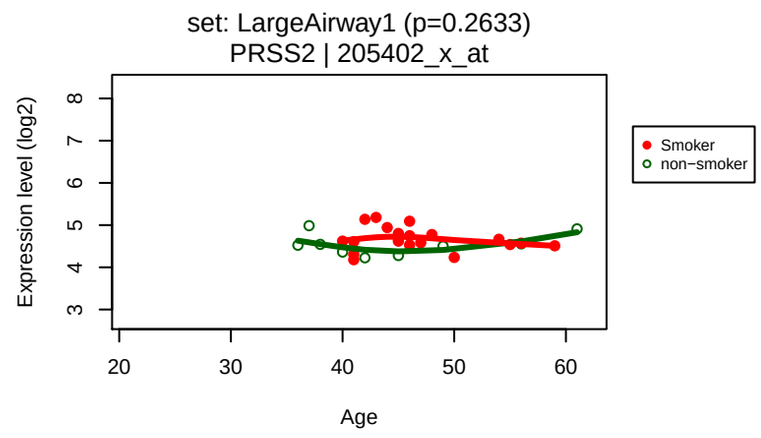

Gene expression of smokers and nonsmokers
gene: PRSS2 | 205402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1394
p=0.2353
p=0.0732
p=0.2633
p=0.0927
p=0.0838
p=0.3313
p=0.8705
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

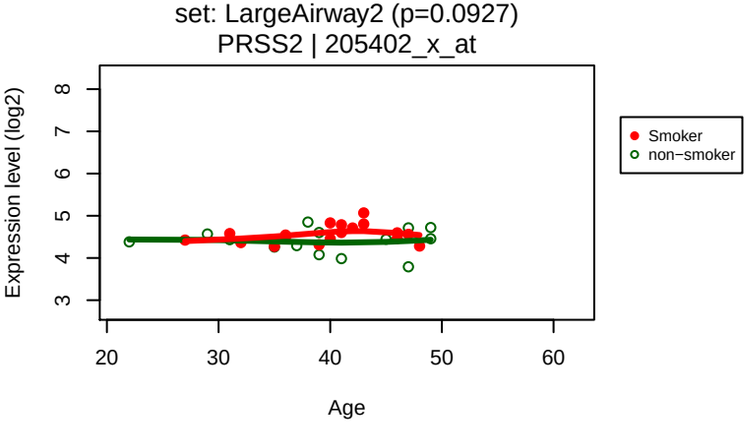

Gene expression ratio (smokers/nonsmokers)
gene: PRSS2 | 205402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1394
p=0.2353
p=0.0732
p=0.2633
p=0.0927
p=0.0838
p=0.3313
p=0.8705
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

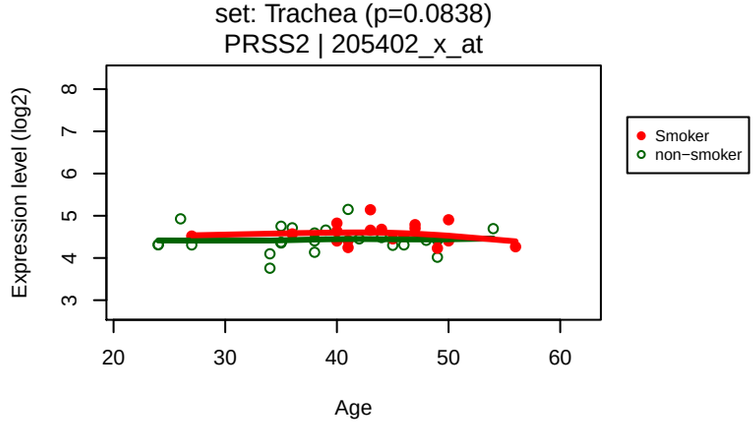

Gene expression ratio in Males
gene: PRSS2 | 205402_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3114
p=0.5966
p=0.1903
p=0.6587
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS2 | 205402_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5338
p=0.0841
p=0.042
p=0.5599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

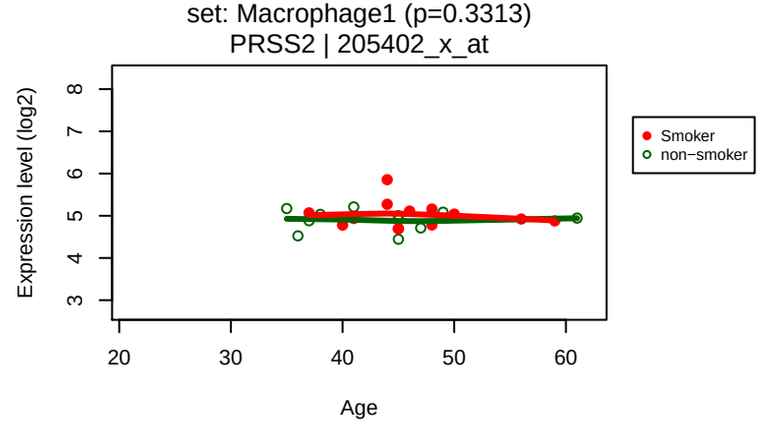

set: SmallAirway1 (p=0.1394)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker

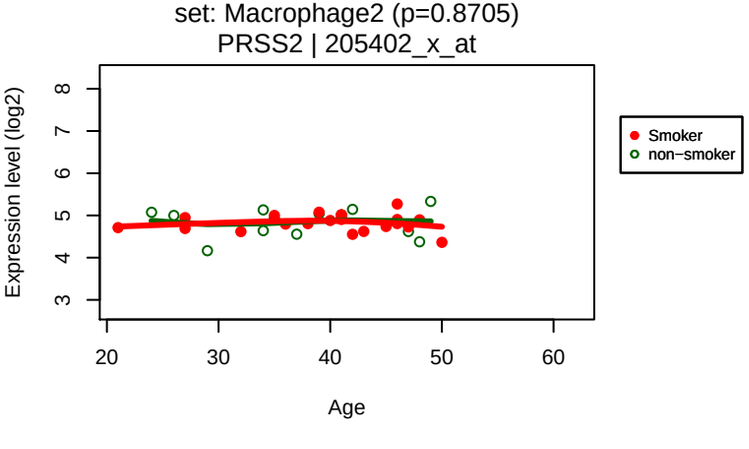

set: SmallAirway2 (p=0.2353)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker

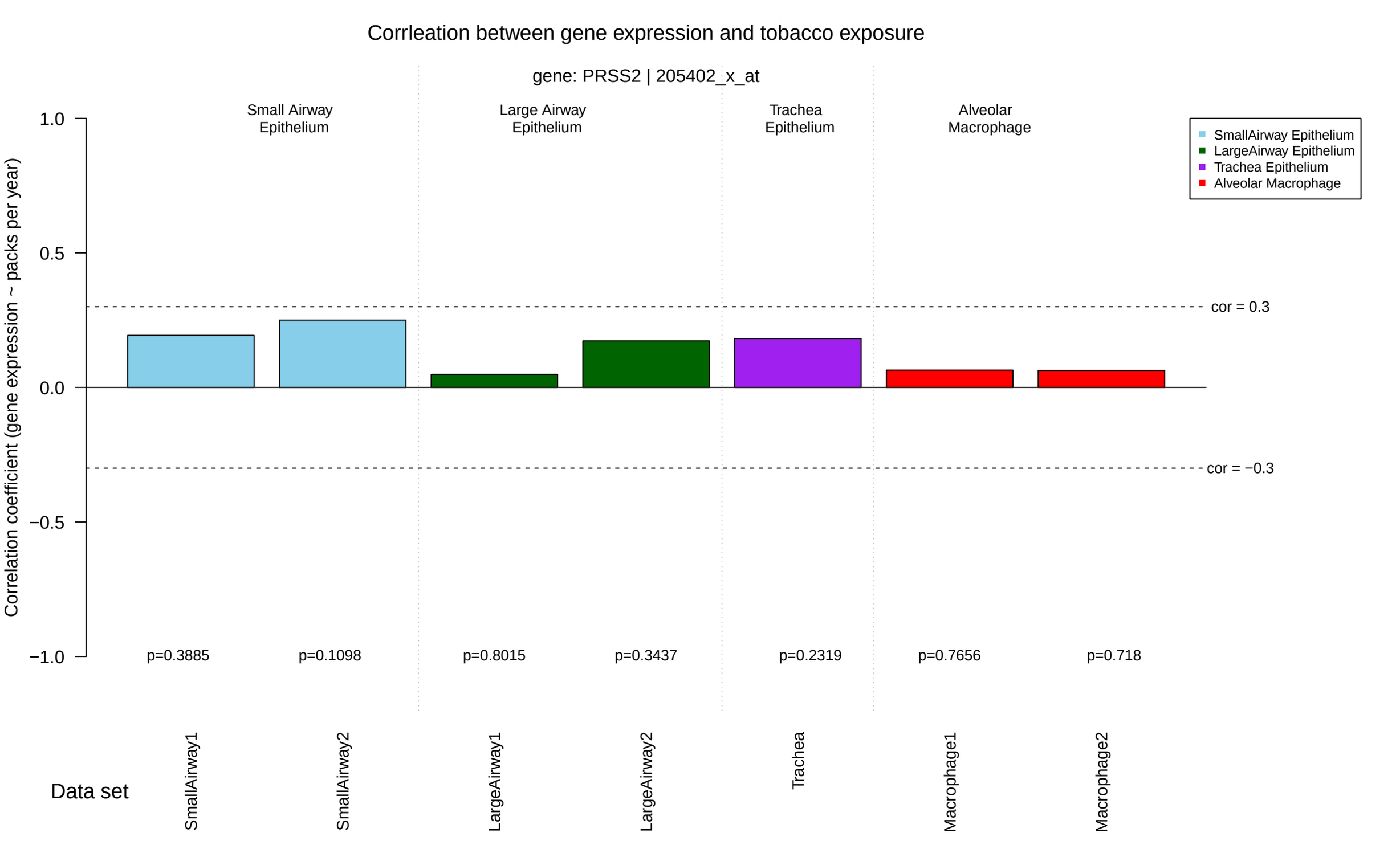

set: LargeAirway1 (p=0.2633)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0927)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0838)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3313)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8705)
PRSS2 | 205402_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS2 | 205402_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3885
p=0.1098
p=0.8015
p=0.3437
p=0.2319
p=0.7656
p=0.718
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2